	AzureSky	
	Hard coal sample preparation procedure	Document no: AS001IW
	Compiled by:	Sifiso Nyambi - Laboratory Manager
	Approved by:	Nonhlanhla Msibi - Operations Manager

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1. INTRODUCTION

This procedure specifies the requirements for the coal sample preparation procedure of proficiency testing scheme and certified reference material. A single sample is obtained from a coal mine or coal laboratory. Constituents of a sample are increments taken from a lot/stockpile.

2. SCOPE

This procedure describes the preparation of samples of coal from a combination of increments representative of the lot.

3. NORMATIVE REFERENCE

ISO 13909-4 Hard coal and coke – Mechanical sampling Part 4: Coal preparation of test samples.

4. SAMPLE PREPARATION

4.1. GENERAL ANALYSIS

1. Request coal sample <200 kg from a mine or coal laboratory.
2. Record the details of the sample onto the receiving book.
3. Air dry the sample until it is visibly dry.

If there are multiple samples of unknown grades, start by collecting small portions (<5kg) from each of the individual samples for ash indicative analysis. Stage crush to 3mm, pulverise and take to the lab. Upon receiving results, the samples shall be grouped according to the following ranges:

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- **10-15 (%) ash**
- **16-20 (%) ash**
- **21-24 (%) ash**
- **25-28 (%) ash**
- **29-32 (%) ash**
- **33-36 (%) ash**
- **37-40 (%) ash**

NB: Discard any samples with an ash >40%.

For samples that come from one stockpile, there is no need to perform an indicative analysis, proceed to crushing after drying.

4. Crush the entire sample in the 25mm crusher.
5. Crush the entire sample in the 10mm crusher.
6. Next, crush the entire sample in the 3mm crusher.
7. Air-dry the sample in an oven ≤40°C for ≤4 hours.
8. Pulverise the entire sample.
9. Transfer the sample into the 210L drum, strap the drum onto the mixer and homogenise for 4 hours.
10. Finally, take increments of ±600g in a pan. Use the flat heap method to take portions from 600g and transfer into ~100g bottles for packaging.
11. In the process take 10 random samples which will be used for homogeneity check and validation of GA parameters.
12. Take about 50 samples for PT and label them. The rest of the samples shall be used as CRMs, and they too should be labelled.

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- **Note: These 10 bottles shall be kept in storage under the same conditions as the CRMs, they will be used to monitor stability of the samples.**
- **It is also worth noting that the top nominal size of all samples received by the lab shall not be >50mm.**

4.2. For Hardgrove and Abrasive index (Eskom Method).

A minimum mass of 300 kg is required:

- Air dry the entire sample, until visibly dry.
- Cone and quarter the entire sample, take two opposites for PT and then cone and quarter the other two opposites. From this take two opposites and add to the first two PT opposites. Approximately 225 kg and 75 kg should be obtained for AI and HGI, respectively.
- **Prepare each of the two parameters separately from this point onwards.**
- Crush the sample at 25mm, 10mm, and 4.75mm in that order (i.e. stage crush).
- ~~Homogenise the sample for 2 hours in a mixing-drum.~~ **(Generates a lot of fines which is not ideal for HGI and AI analysis).** Use cone-and-quartering to reduce the mass of the sample and transfer into the 210L drum ensuring that sample does not cover more than 75% of the area of the drum. Homogenise for 30 minutes.
- Use the -1.0 µm sieve to screen the entire sample (removal of fines). Take the oversize and transfer into a container.
- **For HGI: Weigh 1.0 kg and transfer into a 1L bucket for packaging. Take all the material and keep it in the storage room. Some of the samples will be sent out for packaging while the rest will be stored as CRM.**
- **For AI: Weigh ±9.5 kg and transfer into a 10L bucket for packaging. Send all the samples to the storage room. The required number will be distributed to participants while the rest is kept as CRM.**

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4.3 For Total Moisture (semi-prep).

ISO 589: ***Hard Coal and coke- determination of total moisture***

4.3.1. Sampling

- Obtain a representative sample as per ISO 13909 (<150kg).
- Avoid exposure to air and moisture loss/gain.

4.3.2. Receiving and preparation

- Record the sample in the receiving book.
- Weigh the initial mass.
- Air-dry the sample on the floor, until visibly dry.
- Weigh back the sample.
- Calculate the surface moisture by using the following formula:

  SM = (100- $\frac{Mass\ of\ dry\ sample}{mass\ of\ wet\ sample}$) $\boldsymbol{X\ 100}$

- Crush the sample in the 25mm and 10mm crushers.

4.3.3 Mixing and quartering

- Mix the sample thoroughly to ensure homogeneity.
- Use cone and quartering to divide into test portions
- Crush the sample into the 3mm crusher.
- Quickly weigh 1kg and dry at 105-110 ^{o}C oven.
- Dry the sample for 3 hours.

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The drying depends on the sample size and mass, moisture content in sample and air circulation in the oven

Requirement: The sample should be dried to constant mass (i.e. two successive weighings must be <0.2%, of the sample mass or less than 0.1 g, whichever is greater).

- For PT samples, after reducing them to -3mm weigh ~ 2kg and transfer into 2L buckets. Tightly close the container and attach label. Send to storage room.

Perform Homogeneity Test for HGI, AI, and TM using ash analysis. Take 10 random samples. If the samples do not meet the acceptance criteria, reject the batch and open an IRQ.

5. Reference

- ISO 13909-4
- ISO 589
- Eskom Method

6. Revision table

Revision no	Content changed	Reason for revision	Date changed
01	Homogenise the sample for HGI and AI samples for 2 hours	This step leads to the generation of a lot of fines.	08/08/2025

Effective date	Rev status	Next review date
04/2025	01	04/2028